

Veydrift: A Persistent Onchain Space Economy

Veydrift

veydrift.com

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Abstract

Veydrift is a persistent onchain strategy game in which players settle planets, construct infrastructure, research technologies, manufacture fleets, form alliances, trade, and raid. Three productive resources—metal, crystal, and deuterium—exist both as in-game units and as fully reserve-backed tokens on Base. The Rift connects these domains: market resources enter the game immediately, while resources leaving the game remain locked on a planet for four weeks, visible and vulnerable to raids. A fixed-supply \$VEYDRIFT token routes liquidity between WETH and the three resources. Initial \$VEYDRIFT price discovery uses an onchain continuous clearing auction before liquidity migrates into a Uniswap v4 pool. This paper defines the game formulas, reserve model, token supplies, liquidity structure, Rift mechanics, and economic controls that allow production, conflict, trade, and long-term resource sinks to reinforce one another.

1 Introduction

Veydrift is a persistent onchain strategy game in which players settle planets, build infrastructure, research technologies, manufacture fleets, form alliances, trade, raid, and compete for scarce economic advantage. Its economy is built around three productive resources—metal, crystal, and deuterium—whose generation is deterministic, whose use is extensive, and whose external representation is fully reserve-backed.

The central economic design is deliberately simple:

- **Resources are useful before they are tradable.** Mines create resources; buildings, research, ships, defenses, energy systems, logistics, and recovery consume them.
- **The Rift connects the game to markets.** External resource tokens can enter the game instantly. Leaving the game requires a four-week, planet-bound lock that is visible, unusable, immovable, and raidable.
- **\$VEYDRIFT is the routing and coordination asset.** It is a fixed-supply Base mainnet token paired with WETH. Resource tokens pair only with \$VEYDRIFT, concentrating liquidity and making the game’s economy legible.
- **Launch price discovery is continuous and onchain.** A continuous clearing auction distributes a predeclared tranche over time, reducing first-block timing advantage before proceeds and reserved tokens migrate into the canonical Uniswap v4 pool.
- **The system remains solvent by construction.** Every internal resource liability is backed by the corresponding ERC-20 reserve held by the game contract.
- **Extraction is a strategic event, not a passive yield claim.** A player advertising an exit creates a four-week target, an incentive to defend, and an opportunity for rivals. Security, intelligence, transport, and conflict become economic services.

Each resource token has 6 decimals and a fixed genesis supply of 10 billion whole tokens. Almost all supply begins in the game reserve. One whole token represents one million in-game resource units, so the reserve is backing capacity rather than circulating float.

\$VEYDRIFT has a fixed maximum supply of 1 billion tokens on Base mainnet. Most of the supply not reserved for development, contributors, Pact participants, and strategic uses enters circulation through an auction and protocol-owned liquidity. The launch bootstrap assigns 250 million \$VEYDRIFT to a continuous clearing auction and reserves 250 million for the post-auction \$VEYDRIFT/WETH pool. Before the auction, a 5% Pact round sells 50 million \$VEYDRIFT at \$0.002 per token. The auction floor is \$0.0026 per \$VEYDRIFT, 30% above the Pact price; it is a minimum accepted price, not the expected clearing price or a peg. The three resource pools open at reference prices of \$1.50, \$2.25, and \$3.75 per \$vMETAL, \$vCRYSTAL, and \$vDEUT, encoding the initial 1 : 1.5 : 2.5 scarcity ratio. At the floor price this is approximately:

$$\begin{aligned} 1 \text{ } \$v\text{METAL} &\approx 577 \text{ } \$\text{VEYDRIFT}, & 1 \text{ } \$v\text{CRYSTAL} &\approx 865 \text{ } \$\text{VEYDRIFT}, \\ 1 \text{ } \$v\text{DEUT} &\approx 1,442 \text{ } \$\text{VEYDRIFT}. \end{aligned}$$

Every dollar-denominated amount in this paper is an illustrative estimate derived from the auction floor, discovered clearing price, assumed asset prices, or pool contributions identified alongside it. These figures are neither pegs nor promises of market value, proceeds, yield, or liquidity. Actual WETH, \$VEYDRIFT, \$vMETAL, \$vCRYSTAL, and \$vDEUT prices remain market-discovered and may differ materially.

2 The game economy

2.1 A civilization loop, not a faucet

Every Veydrift account begins with a constrained productive base. A planet creates resources, but meaningful output requires investment in mines and energy. Those resources then fund technologies, ships, defenses, additional planets, logistics, and the infrastructure necessary to support further production. The loop is recursive:

produce → allocate → build → project power → acquire or lose resources → reinvest.

The word “resource” is therefore literal. Metal, crystal, and deuterium are intermediate goods with different productive roles, not interchangeable points. A healthy economy is one in which their marginal utility persists across player maturity.

2.2 Three resources

Resource	Economic role	Designed scarcity
Metal	Bulk construction, hulls, mines, storage, defensive mass, and industrial expansion.	Highest base production; largest absolute sinks.
Crystal	Advanced structures, electronics, research, and higher-technology systems.	Two-thirds of metal's base hourly output; higher reference price.
Deuterium	Energy-intensive systems, propulsion, fleet movement, and strategic logistics.	Lowest base production, planet-temperature sensitivity, and ongoing fuel demand.

The launch price ratio in Section 7 follows the underlying full-power production ratio rather than an arbitrary marketing number. At a neutral planet temperature, base production approaches 30 : 20 : 12.8, suggesting an inverse scarcity ratio near 1 : 1.5 : 2.34. The chosen relative ratio of 1 : 1.5 : 2.5 adds a modest premium for deuterium's logistical utility. Because one external resource token represents one million in-game units, the launch reference prices are \$1.50, \$2.25, and \$3.75 per resource token, converted into \$VEYDRIFT at the discovered auction price.

2.3 Starting state

A standard first planet begins with 500 metal, 500 crystal, and 0 deuterium. A valid referral doubles the starting metal and crystal to 1,000 each. These amounts enter the internal ledger only when the external reserves can back them.

This small genesis grant establishes a critical principle: early progress comes from play and productive decisions, while market imports remain optional acceleration rather than a mandatory admission fee.

3 Production, energy, costs, and time

3.1 Mine production

Let L_M , L_C , and L_D be the levels of the metal, crystal, and deuterium mines. Let μ_P be a planet production multiplier, μ_T the deuterium temperature multiplier, μ_R the crawler multiplier, and μ_E the energy availability multiplier. Current full-power hourly production before reserve constraints is:

$$M_h = \lfloor 30L_M(1.1)^{L_M}\mu_P\mu_R\mu_E \rfloor, \quad (1)$$

$$C_h = \lfloor 20L_C(1.1)^{L_C}\mu_P\mu_R\mu_E \rfloor, \quad (2)$$

$$D_h = \lfloor 10L_D(1.1)^{L_D}\mu_T\mu_R\mu_E \rfloor - D_{\text{fusion}}. \quad (3)$$

For metal and crystal, the standard planet multiplier is $\mu_P = 1$. For deuterium, the basis-point rule is

$$\mu_T = \frac{\max(0, 12800 - 20T)}{10000},$$

where T is planet temperature. At $T = 0$, $\mu_T = 1.28$.

3.2 Crawlers and energy

Each effective crawler adds 0.02% production. The effective crawler count is capped by both infrastructure and a hard aggregate bonus:

$$N_{\text{effective}} = \min(N_{\text{crawler}}, 8(L_M + L_C + L_D)), \quad (4)$$

$$\mu_R = \min(1 + 0.0002N_{\text{effective}}, 1.5). \quad (5)$$

The aggregate 50% ceiling is a safety bound rather than a reachable assumption at current mine limits. With three equal level- L mines, the mine-level cap permits a crawler bonus of $0.48L$ percent: 14.4% at level 30 and 24% at level 50. Economic projections use these attainable caps rather than the aggregate ceiling.

Energy is a real constraint, not flavor text. If demand exceeds production, all mines scale proportionally:

$$\mu_E = \min\left(1, \frac{E_{\text{produced}}}{E_{\text{required}}}\right), \quad E_{\text{required}} = 10L_M + 10L_C + 20L_D$$

after applying the building-specific scaling rules. Fusion generation also consumes deuterium, creating a recurring sink at higher industrial intensity.

3.3 Exponential costs

For a building with base resource vector $\mathbf{b}$, growth factor g , and current level L , the next upgrade cost is

$$\mathbf{C}_{\text{building}}(L) = \lfloor \mathbf{b} g^L \rfloor.$$

Growth factors include 1.5 for the metal mine, 1.6 for the crystal mine, 1.5 for the deuterium mine and solar plant, 1.8 for the fusion reactor, and 2.0 for many advanced structures. Exponential costs are the main long-run sink: production grows at roughly 1.1^L while key upgrade costs grow at 1.5^L or faster. Mature production therefore does not imply unconstrained net extraction.

3.4 Time as a scarce input

Resource costs are only one dimension. Base duration formulas are:

$$t_{\text{building}} \approx \frac{(M + C)3600}{2500(R + 1)2^N s}, \quad (6)$$

$$t_{\text{research}} \approx \frac{(M + C)3600}{1000(L_{\text{lab}} + 1)s}, \quad (7)$$

$$t_{\text{units}} \approx \left\lceil \frac{(M + C)q3600}{2500(S + 1)2^N s} \right\rceil, \quad (8)$$

where R is robotics level, N nanite level, S shipyard level, q quantity, and s the configured speed. Time prevents a liquid market from collapsing all progression into a single instant purchase.

4 Raids, battle resolution, and debris

4.1 Raids transfer player-held resources

A raid is an attack mission against another player's planet. The attacker commits ships, cargo capacity, travel time, and deuterium fuel. The defender contributes stationed ships, planetary defenses, and any allied fleets whose defense or intercept mission is present when the attack arrives. Both sides can lose constructed assets. Only an attacker victory permits plunder; a defender victory or draw returns the surviving attackers without loot.

Ordinary planetary resources are subject to a 50% plunder ceiling. For resource r with planet balance $S_{p,r}$, the ordinary-planet loot ceiling is

$$P_{p,r} = \lfloor 0.5S_{p,r} \rfloor.$$

Actual loot is further bounded by the surviving attacking fleet's unused cargo capacity. Players may choose a desired metal/crystal/deuterium cargo ratio; unused capacity rolls to the other resources when a requested share is unavailable. Loot is credited to the attacking fleet and becomes spendable by the attacker after the surviving fleet returns.

4.2 Combat statistics and rounds

Each ship and defense type has base weapon, shield, and hull values A_0 , S_0 , and H_0 . If its owner has Weapons, Shielding, and Armor technology levels w , s , and a , the effective values are

$$A = A_0(1 + 0.1w), \quad (9)$$

$$S = S_0(1 + 0.1s), \quad (10)$$

$$H = H_0(1 + 0.1a). \quad (11)$$

A battle lasts at most six rounds. Every round begins from a snapshot of the surviving units on both sides, so attacker and defender fire for that round is simultaneous. Shots are distributed across individual surviving units in proportion to unit counts, with deterministic onchain randomness resolving remainders and destruction trials. Catalog rapid-fire relationships can generate additional shots.

For q same-strength shots assigned to a target, let $X = qA$ be incoming damage. A shot whose attack is no more than 1% of the target shield cannot penetrate it. Otherwise shields absorb damage first and the hull damage is

$$d = \max(0, X - S).$$

If $d \geq H$, the target is destroyed. If $0.3H < d < H$, its destruction probability is d/H ; damage of 30% of hull or less does not create an explosion chance. The random stream is committed onchain, making a resolved battle reproducible from public inputs and the finalized random value.

Let N_A and N_D be the surviving attacker and defender unit counts after combat. The outcome is

$$\text{outcome} = \begin{cases} \text{attacker victory,} & N_A > 0 \text{ and } N_D = 0, \\ \text{defender victory,} & N_A = 0 \text{ and } N_D > 0, \\ \text{draw,} & \text{otherwise.} \end{cases}$$

The final case includes mutual destruction and battles in which both sides still have units after six rounds. After combat, 70% of destroyed planetary defenses are repaired. Destroyed ships are permanent losses.

4.3 Debris fields

Destroyed ships from both sides create a debris field at the attacked planet. Let L_M^A, L_C^A and L_M^D, L_C^D be the metal and crystal construction value of attacker and defender ship losses. The added debris is

$$D_M = \lfloor 0.30(L_M^A + L_M^D) \rfloor, \quad (12)$$

$$D_C = \lfloor 0.30(L_C^A + L_C^D) \rfloor, \quad (13)$$

$$D_D = 0. \quad (14)$$

Defense losses create no debris. Debris persists at the coordinate, accumulates across battles, and can be collected by a harvest fleet up to its remaining cargo capacity. When both resources are present, harvesting takes from both sides of the field before using remaining capacity on the larger balance. Every 100,000 combined metal and crystal debris creates a 1% moon-formation chance, capped at 20% for a single battle.

5 Resource-token architecture

5.1 Supply and denomination

Asset	Symbol	Decimals	Fixed supply
Veydrift Metal	\$vMETAL	6	10,000,000,000
Veydrift Crystal	\$vCRYSTAL	6	10,000,000,000
Veydrift Deuterium	\$vDEUT	6	10,000,000,000

The deployed token symbols are \$vMETAL, \$vCRYSTAL, and \$vDEUT. One whole external resource token equals 1,000,000 in-game resource units. The token quantity and the user-interface resource quantity must never be conflated:

$$\begin{aligned} 1 \text{ \$vMETAL} &= 10^6 \text{ metal units,} & 1 \text{ \$vCRYSTAL} &= 10^6 \text{ crystal units,} \\ 1 \text{ \$vDEUT} &= 10^6 \text{ deuterium units.} \end{aligned}$$

5.2 Reserve invariant

For resource r , define B_r as the resource-token balance held by the game contract and I_r as total internal resource liabilities. Solvency requires

$$B_r \geq I_r.$$

The protocol blocks new internal issuance when the corresponding external reserve is insufficient. Market deposits increase both B_r and I_r by the same amount; finalized withdrawals decrease both by the same amount. This is a transparent, auditable reserve system rather than an algorithmic peg.

5.3 Contract control and supply policy

Each resource token has a hard supply cap of 10 billion. Resource-market liquidity is released from provable excess reserve rather than created through additional minting. Supply policy follows three rules:

1. The initial AMM float is released from the existing 10-billion-token reserve only when the remaining reserve fully covers internal liabilities.
2. No additional resource tokens are minted for launch liquidity.
3. Reserve, internal liability, locked exit, and circulating float are reported separately for each resource.

6 The \$VEYDRIFT token

6.1 Purpose

\$VEYDRIFT coordinates the economy without replacing the game. Its base functions are deliberately narrow:

- the sole routing asset for resource liquidity;
- the WETH-facing liquidity and price-discovery asset;
- the common quote and settlement asset between the three resource markets.

The same token can later support optional uses such as market fees, alliance services, cosmetics, tournaments, ecosystem grants, buybacks, or burns. None of those extensions is required by the base economic model. \$VEYDRIFT does not sell hidden combat advantages or bypass build times; resources remain the explicit market for economic acceleration.

6.2 Genesis parameters

Parameter	Value or assumption
Name / symbol	Veydrift / \$VEYDRIFT
Network	Base mainnet
Decimals	18
Maximum supply	1,000,000,000 \$VEYDRIFT
Inflation	None after genesis
Pact price	\$0.002 per \$VEYDRIFT (\$2,000,000 FDV)
CCA floor price	\$0.0026 per \$VEYDRIFT, 30% above the Pact price (minimum, not a target)
Floor FDV	\$2,600,000 at the floor price
CCA minimum raise	About \$150,000 of WETH; below it the auction does not graduate and bids are refunded

The full supply is created at genesis, and the economic model assumes no further issuance. The CCA floor is used to model minimum launch values; it is not a peg, target, or promise that the token will trade at that value. The floor is fixed in WETH at a reference ETH price shortly before the auction, so its dollar value moves with ETH.

6.3 Allocation

Allocation	Share	Tokens	Release structure
CCA public distribution	25%	250,000,000	Distributed through the Base CCA over a 48-hour window.
Uniswap v4 main liquidity	25%	250,000,000	Reserved for automatic migration into the canonical \$VEYDRIFT/WETH pool.
Initial resource liquidity	15%	150,000,000	Circulating at launch; 50,000,000 tokens pair with each resource.
Development reserve	15%	150,000,000	Five-year linear release for development, infrastructure, security, and operations.
Core contributors	5%	50,000,000	Four-year vesting with a one-year cliff. Unsold Pact tokens remain here.
Pact	5%	50,000,000	Sold before the auction at \$0.002 per token. 12.5% releases six months after the auction ends; the rest releases linearly over the following 3.5 years.
Ecosystem / strategic	10%	100,000,000	Six-year release for integrations, partnerships, and market expansion. Unsold auction tokens are added here.
Total	100%	1,000,000,000	

Initial circulation. The auction, post-auction main pool, and three resource pools together place 650,000,000 \$VEYDRIFT, or 65% of maximum supply, into launch circulation. At the \$0.0026 floor, the corresponding floor-value arithmetic is \$1,690,000; the auction may discover a higher price. This is scenario arithmetic, not a promised valuation. The remaining 350,000,000 tokens are reserved for development, contributors, Pact participants, and ecosystem or strategic uses and do not count as

circulating until their releases become transferable.

Release discipline. Auction and launch-liquidity allocations are fixed before bidding and are not supported by later inflation. Development, contributor, Pact, and strategic allocations follow their release schedules without accelerating because of short-term market conditions. Unsold auction tokens join the ecosystem allocation’s six-year release, and migration residuals remain restricted to the published launch-liquidity mandate; neither becomes discretionary contributor inventory.

7 Liquidity bootstrap and price discovery

7.1 Pool topology

The liquidity graph is intentionally sparse:

$$\text{WETH} \longleftrightarrow \$\text{VEYDRIFT} \longleftrightarrow \{\$v\text{METAL}, \$v\text{CRYSTAL}, \$v\text{DEUT}\}.$$

There are no protocol-seeded resource/WETH or resource/stablecoin pools. This concentrates price discovery, increases the utility of \$VEYDRIFT, and avoids fragmenting thin resource liquidity. The trade-off is that resource pricing inherits \$VEYDRIFT volatility and most resource-to-WETH trades use two hops. A production router must quote both hops, enforce user-defined slippage, and use time-weighted prices only for protocol accounting.

7.2 Continuous clearing auction

The launch begins with Uniswap’s Continuous Clearing Auction (CCA) on Base rather than an immediately tradable fixed-price AMM. The auction tranche is 250,000,000 \$VEYDRIFT, the quote currency is WETH, and the duration is 48 hours expressed as immutable start and end block numbers. The floor is set 30% above the Pact price and the minimum raise is about \$150,000 of WETH; if the auction raises less, it does not graduate and bidders are refunded. Exact blocks, the WETH-denominated floor, minimum raise, migration block, claim block, fee settings, recipients, and contract addresses are published before bidding begins.

Each bid declares a WETH budget and maximum accepted \$VEYDRIFT price. The protocol spreads the bid across the auction’s remaining blocks. Within each block, successful bidders pay one clearing price; higher maximum-price bids fill first and marginal bids fill pro rata. Supply is released gradually, so demand cannot be concentrated into a single first or final transaction. Early participation receives exposure to more clearing blocks, but transaction ordering alone does not grant the entire launch allocation.

The CCA reduces first-block and last-minute timing games; it does not eliminate automation, Sybil participation, market risk, or price volatility. There is no owner-controlled ERC-20 transfer tax, blacklist, maximum-wallet rule, or hidden trading switch. Auction parameters are immutable after creation. The production launch uses the current audited CCA deployment supported by the Uniswap Liquidity Launchpad on Base and publishes the exact deployed version.

After the end block, the auction is finalized. If its predeclared graduation condition is met, the liquidity strategy initializes the canonical Uniswap v4 \$VEYDRIFT/WETH pool at the discovered price. Auction

proceeds and the 250,000,000-token liquidity reserve supply the pool, subject to the published position plan and currency cap. Unsold auction tokens go to the ecosystem allocation; any other residual token or WETH follows the predeclared strategy into disclosed liquidity positions or the launch Safe after the sweep delay. Participants claim purchased tokens only from the auction contract after the published claim block.

7.3 Initial liquidity

At the \$0.0026 floor, a fully sold 250,000,000-token auction tranche raises an estimated \$650,000 of WETH-equivalent value before fees. The 250,000,000-token migration reserve has the same \$650,000 floor value, implying approximately \$1,300,000 of main-pool depth if the tranche clears at the floor and the position plan uses the balanced amounts. If the auction raises only its \$150,000 minimum at the floor, about 57.7 million tokens sell and the main pool starts near \$300,000 of depth. A higher clearing price increases the WETH raised and changes every dollar estimate. The contracts use token quantities and WETH amounts, not a dollar oracle.

Component	External side	\$VEYDRIFT side	Pricing method	Floor-value estimate
CCA distribution	WETH bids	250,000,000	Per-block clearing	\$650,000 distributed value
\$VEYDRIFT/WETH LP	Auction WETH	250,000,000	Discovered CCA price	\$1,300,000 pool depth
\$vMETAL/\$VEYDRIFT	86,666.667	50,000,000	\$1.50 reference	\$260,000 est.
\$vCRYSTAL/\$VEYDRIFT	57,777.778	50,000,000	\$2.25 reference	\$260,000 est.
\$vDEUT/\$VEYDRIFT	34,666.667	50,000,000	\$3.75 reference	\$260,000 est.
Launch circulation		650,000,000 \$VEYDRIFT		\$2,080,000 DEX depth est.

The resource pools are initialized only after the main pool establishes the \$VEYDRIFT/WETH price. Their \$VEYDRIFT-per-resource ratios are set from the discovered price so that each resource opens at its reference price while preserving the intended 1 : 1.5 : 2.5 relative scarcity. A higher clearing price pairs more resource tokens with the same 50,000,000 \$VEYDRIFT per pool; resource amounts in the table assume the floor. Resource tranches are released from excess game reserves only when post-transfer backing remains greater than internal liabilities plus an operational safety margin. Market depth can expand with actual demand.

8 Production valuation and mature-production stress model

8.1 Level-25 valuation anchor

The resource-price scenario is anchored to a developed planet with level-25 metal, crystal, and deuterium mines. At full energy, neutral temperature, no fusion consumption, and the maximum crawler count effective at those mine levels, that planet produces 218,424 metal, 145,608 crystal, and 93,144 deuterium per day. Because one external token represents one million in-game units, the valuation model uses the launch reference prices:

$$\begin{aligned}
P_M &= \$1.50 \text{ per } \$v\text{METAL (launch reference),} \\
P_C &= \$2.25 \text{ per } \$v\text{CRYSTAL (launch reference),} \\
P_D &= \$3.75 \text{ per } \$v\text{DEUT (launch reference).}
\end{aligned}$$

Under those price assumptions, the estimated gross reference value is

$$0.218424P_M + 0.145608P_C + 0.093144P_D \approx \$1.00 \text{ per day (estimate)}.$$

This estimate is not a yield promise. It depends entirely on the assumed token prices as well as energy availability, planet temperature, crawler ownership, construction spending, fuel, raids, the four-week Rift exposure, liquidity, and AMM price impact. Actual realizable value may be lower, higher, or unavailable.

8.2 Reference production by mine level

For mature-production stress testing, the model assumes equal mine levels, full energy, the maximum crawler count attainable at that mine level, planet temperature $T = 0$, no fusion consumption, no raids, no downtime, and no reinvestment. The dollar estimates continue to assume \$1.50 per \$vMETAL, \$2.25 per \$vCRYSTAL, and \$3.75 per \$vDEUT; none is a promised market price.

Mine level	M / h	C / h	D / h	M / day	C / day	D / day	Est. value / day
25	9,101	6,067	3,881	218,424	145,608	93,144	\$1.00 est.
30	17,965	11,976	7,663	431,160	287,424	183,912	\$1.98 est.
35	34,464	22,975	14,702	827,136	551,400	352,848	\$3.81 est.
40	64,738	43,158	27,619	1,553,712	1,035,792	662,856	\$7.14 est.
45	119,656	79,770	51,052	2,871,744	1,914,480	1,225,248	\$13.21 est.
50	218,346	145,563	93,159	5,240,304	3,493,512	2,235,816	\$24.10 est.

Because each external token represents one million game units, a level-30 planet produces approximately 0.431 \$vMETAL, 0.287 \$vCRYSTAL, and 0.184 \$vDEUT per day at the attainable crawler cap. At the assumed token prices, its estimated combined gross resource value is \$1.98 per day. A level-50 planet produces approximately 5.24 \$vMETAL, 3.49 \$vCRYSTAL, and 2.24 \$vDEUT, with an estimated combined gross value of \$24.10 per day under the same assumptions. Neither estimate promises a realizable market value; both precede construction and research spending, crawler acquisition costs, fuel consumption, attacks, the four-week exit lock, liquidity, and market impact.

8.3 Mature multi-planet extreme

The planet-cap formula is one base planet plus Astrophysics level. For scenario analysis, 51 level-50 planets represent an extreme account:

$$M_{30d} \approx 8,018 \text{ $vMETAL,} \quad (15)$$

$$C_{30d} \approx 5,345 \text{ $vCRYSTAL,} \quad (16)$$

$$D_{30d} \approx 3,421 \text{ $vDEUT.} \quad (17)$$

A single level-30 planet's daily output equals approximately 0.129%, 0.129%, and 0.138% of the opening metal, crystal, and deuterium pool reserves. Thirty days of uninterrupted output is approximately 3.9%, 3.9%, and 4.1% of those reserves. The opening pools can therefore support early production and price discovery, but they are not intended to absorb a mature population at their initial size.

At the assumed resource-token prices, the 51-planet level-50 scenario has an estimated gross value of about \$1,229 per day, not a promised return. Its 30-day output exceeds the opening reserve of every resource pool. This adversarial bound makes the scaling requirement explicit. The Rift does not impose a quantity limit or daily completion quota: every surviving amount can exit after its lock. Large sales instead move the AMM price directly. The four-week public lock and full raid exposure create advance notice and risk, while progression sinks and deeper market-supplied liquidity determine how much mature production can be sold near the assumed reference prices.

8.4 The sink side dominates late upgrades

At the protocol formulas, selected level-50 next-upgrade costs are enormous:

Upgrade from level 50	Metal cost	Crystal cost	Observation
Metal mine	38,257,290,012	9,564,322,503	More than 6,000 days of one level-50 planet's metal output before other inputs.
Crystal mine	771,330,261,244	385,665,130,622	Cost growth substantially exceeds production growth.
Deuterium mine	143,464,837,548	47,821,612,516	Sustains major metal/crystal demand even for deuterium specialists.

The game therefore has a natural sink ladder. The design goal is not to force every player to pursue theoretical maximum levels; it is to keep attractive productive, military, logistical, and social uses available at every maturity band.

9 The Rift: market entry and strategic exit

9.1 Requirements and expected progression

The Rift is unlocked separately on each planet by constructing one Rift Stabilizer. Its direct requirements are Robotics Factory level 4, Research Lab level 2, Energy Technology level 5, Hyperspace Technology level 1, and the one-time Stabilizer cost of 8,000 metal, 8,000 crystal, and 4,000 deuterium.

Hyperspace Technology adds important nested requirements: Research Lab level 7 and Shielding Technology level 5, while Shielding itself requires Energy Technology level 3 and Research Lab level 6. The effective unlock path is therefore

Robotics 4 + Research Lab 7 + Energy 5 + Shielding 5 + Hyperspace 1 + Rift Stabilizer.

A deterministic no-import benchmark using the live costs, storage limits, energy model, and separate construction and research queues reaches a first Rift in approximately 14 days. The benchmark keeps queues active, develops roughly 12/11/9 mines with sufficient solar power, and spends only on required production and unlock infrastructure. A player who also builds fleets, defenses, or unrelated research should expect roughly two to three weeks. Raids, idle queues, energy shortages, and resource losses extend the schedule; market imports can fund the costs but do not remove construction and research time.

9.2 Instant market-to-game entry

A market deposit transfers resource tokens to the game contract and immediately credits the selected planet. Imported resources already exist externally and increase reserve and internal liability together.

If a user deposits q units of resource r :

$$B'_r = B_r + q, \quad I'_r = I_r + q.$$

The reserve invariant is unchanged.

9.3 Four-week planet lock

Resources leave the game through a 28-day, planet-scoped Rift lock. Let $S_{p,r}$ be spendable resource r on planet p and $Q_{p,r}$ its exit-locked amount. A request of q performs

$$S'_{p,r} = S_{p,r} - q, \quad Q'_{p,r} = Q_{p,r} + q.$$

There is no protocol quantity cap and no daily export-throughput limit. The total internal liability does not change when the lock begins. For 28 days, $Q_{p,r}$:

- cannot be spent, moved, traded internally, relocked, or used as collateral;
- cannot be withdrawn early or cancelled to evade an incoming attack;
- remains visibly associated with the planet and its unlock time;
- is included in raidable resources under explicit, auditable rules;
- creates a clear target signal for rivals and a defense problem for the owner.

When another player raids locked resources, the attacker's internal balance rises and the original withdrawal claim falls one-for-one; aggregate internal liability remains unchanged. At maturity, every surviving locked unit can be ejected:

$$Q'_{p,r} = Q_{p,r} - q_{\text{surviving}}, \quad B'_r = B_r - q_{\text{surviving}}, \quad I'_r = I_r - q_{\text{surviving}}.$$

The lock is evaluated by contract timestamps and state, not a trusted server. The user interface displays a countdown, the original request, current surviving claim, raid losses, and final ejection amount.

9.4 Raid semantics

Rift balances deliberately use different protection rules from ordinary planet balances. Ordinary planet resources are subject to the 50% plunder ceiling and may be shielded by newbie protection or anti-bashing rules. Rift-locked resources have no newbie protection, no anti-bashing protection, and a 100% plunder ceiling. The public lock is therefore a fully contestable balance, not a protected withdrawal queue.

After an attacker wins, surviving cargo capacity is filled in two stages. First it takes ordinary planet resources, up to the 50% ceiling. Only after the available ordinary-planet tranche is exhausted does remaining cargo take Rift resources, up to 100% of the locked balance:

$$0 \leq L_{p,r}^P \leq \lfloor 0.5S_{p,r} \rfloor, \quad (18)$$

$$0 \leq L_{p,r}^Q \leq Q_{p,r}, \quad (19)$$

$$\sum_r (L_{p,r}^P + L_{p,r}^Q) \leq K, \quad (20)$$

where K is the surviving fleet's unused cargo capacity. The ordering is strict: planet loot first, Rift loot second. A successful raid reduces the owner's corresponding Rift claim and moves the looted resources into the attacker's returning fleet. Losing the battle yields neither tranche.

10 Why the economy can prosper

10.1 A circular demand system

The economy prospers when every participant has a reason to transact with another participant:

1. Builders import or acquire resources to accelerate productive infrastructure.
2. Infrastructure creates resources but also opens exponentially larger sinks.
3. Fleet operators demand deuterium for movement and all resources for replacement capacity.
4. Exporters create visible four-week targets.
5. Raiders demand intelligence, timing, cargo, fleet strength, and fuel.
6. Exporters demand defenses, alliance protection, counterintelligence, and deterrence.
7. Market activity generates exchange fees for liquidity providers and supports deeper price discovery.
8. Alliances can specialize in protection, trade, logistics, retaliation, and collective infrastructure.

The Rift is thus not merely a bridge. It converts a financial action into map-level content. An attempted exit can create more gameplay than an entry because the exit announces value, location, duration, and opportunity.

10.2 No guaranteed yield

Production is not a guaranteed return; it is an input to a competitive game economy. Its external value depends on player demand, sink utility, conflict, liquidity, and market conditions. An economy that treats mine output as passive yield attracts extractors faster than players and eventually consumes its own demand.

The system instead provides deterministic game rules, transparent reserves, credible supply controls, open markets, and a world in which economic decisions create strategic consequences.

11 Risk analysis and controls

Risk			Failure mode	Control
Resource integrity	supply in-		Unexpected issuance dilutes float or reserve credibility.	Fixed 10B cap, release float from existing reserve, onchain alerts, and a public reserve dashboard.
Aggregate wave	export		Many mature accounts sell simultaneously into shallow pools.	Four-week public locks reveal pending supply, all locked resources remain fully raidable, and AMM price impact makes large immediate sales progressively less attractive. There is no export quota.
Two-hop volatility			Resource prices inherit \$VEYDRIFT/WETH volatility and compound slippage.	Deepest pool is \$VEYDRIFT/WETH; router quotes both hops; user-set slippage; TWAP for protocol accounting; no thin direct pools seeded by protocol.
Auction configuration			A bad floor, duration, supply schedule, recipient, or migration plan produces weak distribution or failed migration.	Immutable parameters disclosed before bidding, independent review, Base-fork rehearsal, minimum-raise graduation condition, delayed sweep, and exact post-auction balance assertions.
Pact sell pressure			Early buyers sell into thin post-auction liquidity.	Pact tokens are locked until six months after the auction ends; 12.5% releases then and the rest releases linearly over 3.5 years. The auction floor sits 30% above the Pact price.
Launch automation / sniping			Automated bidders seek ordering or information advantages.	Continuous per-block clearing, bids spread across remaining blocks, one clearing price per block, no first-block AMM, and transparent onchain parameters. CCA reduces timing games but does not promise bot-free distribution.
Impermanent loss			LP providers lose relative value during strong directional moves.	Honest disclosure, protocol-owned liquidity, measured liquidity expansion, and no guaranteed APY.
Sybil / automation			Automated accounts attempt to industrialize farming or extraction.	A strong economy and gameplay mechanics make automation bear the same costs and risks as other players; bot planets remain visible, raidable farming targets rather than privileged faucets.

Risk	Failure mode	Control
Raid concentration	Dominant alliances suppress smaller exporters.	Fleet replacement costs, fuel, public combat rules, intelligence uncertainty, counter-raids, alliance defense, and unrestricted targeting of valuable Rift balances create economic counterpressure.
Contract upgrade risk	Contract changes alter reserve, lock, or raid rules unexpectedly.	Every upgrade and its resulting byte-code are public and onchain, allowing users to inspect the exact change and its execution history.

12 Metrics that determine whether the design works

The protocol publishes the following metrics by resource:

- reserve coverage B_r/I_r and absolute safety margin;
- internal spendable resources, exit-locked resources, and external circulating float;
- 30-day production, consumption, imports, export requests, raid losses, and completed exports;
- resource velocity and the share of production reinvested in sinks;
- AMM depth, volume, fee revenue, LP concentration, and price impact for 30 days of modeled mature production;
- CCA bidder count, wallet concentration, fill distribution, clearing-price path, WETH raised, and migration residuals;
- distribution of mine levels, planets, fleet power, and export activity;
- the fraction of locked exits attacked and the value successfully defended;
- \$VEYDRIFT circulating supply, scheduled unlocks, treasury inventory, and protocol-owned liquidity.

Parameter tuning responds to data, while core property rights—fixed resource supply, reserve backing, lock duration, and protection from arbitrary seizure—remain unchanged.

13 Launch structure

\$VEYDRIFT launches on Base mainnet with a fixed 1-billion-token supply created once. Source code, allocations, vesting contracts, auction parameters, strategy contracts, and treasury addresses are public.

Before the auction, the Pact sells 50,000,000 \$VEYDRIFT (5% of supply, taken from the contributor allocation) at \$0.002 per token, raising up to \$100,000. Pact proceeds fund game improvements and marketing to grow the player base before the auction. Pact tokens are locked until six months after the auction ends, when 12.5% releases; the remainder releases linearly over the following 3.5 years.

At the announced launch block, a 48-hour CCA opens for 250,000,000 \$VEYDRIFT. After the auction ends and graduates, its proceeds and the separate 250,000,000-token liquidity reserve migrate into the canonical Uniswap v4 \$VEYDRIFT/WETH pool at the discovered price. The three resource/\$VEYDRIFT

pools are then initialized from that price. Resource float is released from excess reserve without new minting. Protocol LP positions, lock terms, pool parameters, auction receipts, migration residuals, and treasury movements remain publicly auditable.

The production runbook treats auction creation, token funding, finalization, migration, resource-pool initialization, LP custody, and approval revocation as one rehearsed sequence with explicit abort conditions. No duplicate protocol-seeded \$VEYDRIFT/WETH pool launches on another DEX. A later secondary venue requires separate depth, disclosure, and treasury approval rather than silently fragmenting the canonical market.

Rift exports have no quantity ceiling or daily completion limit. Contract changes and the bytecode they install are public and onchain. Fixed token supplies, reserve backing, active lock state, and user balances remain directly auditable.

14 Conclusion

Veydrift can support an open economy because the economic layer reinforces the game rather than floating above it. Resources are generated by infrastructure, consumed by ambition, moved by logistics, contested by fleets, and backed by verifiable reserves. \$VEYDRIFT can concentrate liquidity and coordination without pretending that a token is the game itself.

The four-week Rift exit is the defining mechanism. Imports are immediate because they add backing. Exports are slow because removing value creates strategic notice, risk, and content. A player who wants to move resources out of the universe must first defend them inside it.

The result is a system in which markets can deepen play: builders bring demand, producers create supply, sinks preserve utility, raiders challenge extraction, alliances sell security, LPs enable exchange, and transparent contracts protect the rules. Prosperity is not guaranteed by emissions. It is earned through useful assets, credible constraints, and persistent competition.

15 Protocol addresses

Game contract	0xf397910F005151b09644228573a4353818D3755d
Metal token	0x91A4f8A9D05F21E010dc1eE0B17Ab644D433cB41
Crystal token	0xC6881a2C4C50E28AdCaC4D5577cD8e211E806B76
Deuterium token	0x5A6027DE1C7E52B4b1AD0c13c3eC3Ad5FCb481e2
